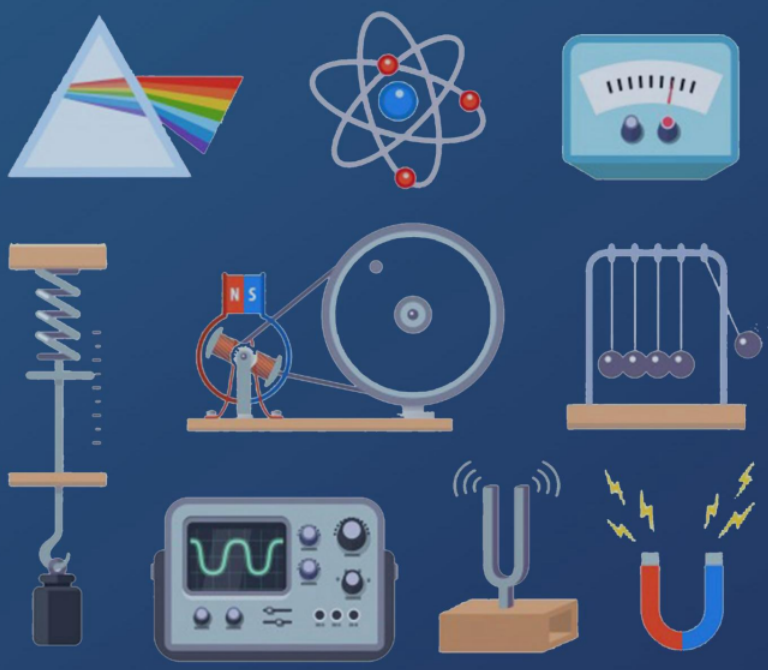
PHYSICS

Electromagnetic Waves



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-2
1	Displacement current	1-3	1-3
2	Electromagnetic wave	4-11	4-10
3	Energy in EM waves	12,13	11-13
4	Electromagnetic Spectrum	14,15	

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Displacement current

Q1. A large parallel plate capacitor, whose plates have an area of 1 m^2 and are separated from each other by 1 mm, is being charged at a rate of 25.8 V/s. If the dielectric between the plates has the dielectric constant 10, then the displacement current at this instant is :-

(A) 251 μA (B) 11 μA
(C) 2.2 μA (D) 1.1 μA

Q2. The conduction current is the same as displacement current when source is :-

(A) ac only (B) dc only
(C) both ac and dc (D) neither dc nor ac

Directions for Assertion & Reason Questions

These questions Consist of two Statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

(A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(C) If Assertion is True but the Reason is false.
(D) If both Assertion & Reason are false.

Q3. The S.I unit of displacement current is-

(A) Henry (B) Coulomb
(C) Ampere (D) Farad

Electromagnetic wave

Directions for Assertion & Reason Questions

These questions Consist of two Statements each, printed as Assertion and Reason. While answering these Questions you are required to choose any one of the following four responses.

(A) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(B) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(C) If Assertion is True but the Reason is false.
(D) If both Assertion & Reason are false.

Q4. **Assertion :-** Electromagnetic radiations exert pressure.

Reason :- EM-waves carry both momentum & energy.

(A) A (B) B
(C) C (D) D

Q5. The electric field associated with an e.m. wave in vacuum is given by

$\vec{E} = 40 \cos (kz - 6 \times 10^8 t)\hat{i}$,

where E, z and t are in volt/m, meter and seconds respectively. The value of wave vector k is:

(A) 6 m^{-1} (B) 3 m^{-1}
(C) 2 m^{-1} (D) 0.5 m^{-1}

Q6. A plane electromagnetic wave of frequency 40 MHz travels in free space in the X-direction. At some point and at some instant, the electric field $\vec{E}$ has its maximum value of 750 N/C in Y-direction. The wavelength of the wave is –

(A) 3.5 m (B) 5.5 m
(C) 7.5 m (D) 9.5 m

Q7. The speed of electromagnetic radiation in vacuum is :-

(A) $\mu_0 \epsilon_0$ (B) $\sqrt{\mu_0 \epsilon_0}$

(C) $\frac{1}{\mu_0 \epsilon_0}$ (D) $\frac{1}{\sqrt{\mu_0 \epsilon_0}}$

Q8. The nature of electromagnetic wave is –

(A) Longitudinal

(B) Longitudinal stationary

(C) Transverse

(D) Transverse stationary

Q9. A plane electromagnetic wave, $E_z = 100 \cos (6 \times 10^8 t + 4x)$ V/m propagates in a medium of dielectric constant :

(A) 1.5 (B) 2.0 (C) 2.4 (D) 4.0

Q10. A beam of light travelling along x-axis is described by the electric field,

$E_y = (600\ Vm^{-1}) \sin \omega (t - x/c)$

then maximum magnetic force on a charge q = 2e, moving along y-axis with a speed of $3.0 \times 10^7\ ms^{-1}$ is ($e = 1.6 \times 10^{-19}$ C) :

(A) 19.2×10^{-17} N (B) 1.92×10^{-17} N

(C) 0.192 N (D) None ofthese

Q11. A plane electromagnetic wave travelling along the X-direction has a wavelength of 3mm. The variation in the electric field occurs in the Y-direction with an amplitude $66 Vm^{-1}$.The equation for the electric and magnetic fields as a function of x and t are respectively.

(A) $E_y = 33 \cos \pi \times 10^{11} \left(t - \frac{x}{c}\right)$;

$B_z = 1.1 \times 10^{-7} \cos \pi \times 10^{11} \left(t - \frac{x}{c}\right)$

(B) $E_y = 11 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$;

$B_y = 1.1 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$

(C) $E_x = 33 \cos \pi \times 10^{11} \left(t - \frac{x}{c}\right)$;

$B_x = 1.1 \times 10^{-7} \cos \pi \times 10^{11} \left(t - \frac{x}{c}\right)$

(D) $E_y = 66 \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$;

$B_z = 2.2 \times 10^{-7} \cos 2\pi \times 10^{11} \left(t - \frac{x}{c}\right)$

Energy in EM waves

Q12. In an electromagnetic wave the average energy density is associated with –

(A) Electric field only

(B) Magnetic field only

(C) Equally with electric and magnetic fields

(D) Average energy density is zero

Q13. In the above problem, the average energy density associated with the electric field will be –

(A) $\frac{1}{2} CV^2$ (B) $\frac{1}{2} \frac{q^2}{C}$

(C) $\frac{1}{2} \frac{\varepsilon^2}{E}$ (D) $\frac{1}{2} \varepsilon_0 E^2$

Electromagnetic Spectrum

Q14. The frequency from 3×10^9 Hz to 3×10^{10} Hz is –

(A) High frequency band

(B) Super high frequency band

(C) Ultra high frequency band

(D) Very high frequency band

Q15. The frequency from 3 to 30 MHz is known as –

(A) Audio band

(B) Medium frequency band

(C) Very high frequency band

(D) High frequency band

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Instantaneous displacement current of 1.0A in the space between the parallel plates of 1μF capacitor can be established by changing potential difference of :

(A) 10^{-6} V/s (B) 10^{6} V/s

(C) 10^{-8} V/s (D) 10^{8} V/s

Q2. The maxwell's equation :

$\oint \vec{B}.\vec{dl} = \mu_0\left(i+\varepsilon_0.\frac{d\phi_E}{dt}\right)$ is a statement of-

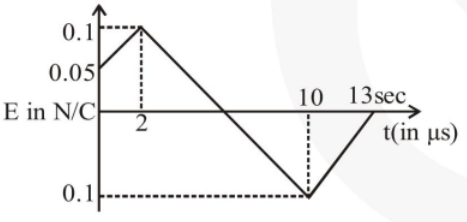
(A) Faraday's law of induction

(B) Modified Ampere's law

(C) Gauss's law of electricity

(D) Gauss's law of magnetism

Q3. The electric field through an area of $2m^2$ varies with time as shown in the graph. The greatest displacement current through the area is at :-



(A) t = 1 sec. (B) t = 4 sec.

(C) t = 8 sec. (D) t = 12 sec.

Q4. If $\vec{E}$ and $\vec{B}$ are the electric and magnetic field vectors of electromagnetic waves then the direction of propagation of electro-magnetic wave is along the direction of –

(A) $\vec{E}$ (B) $\vec{B}$

(C) $\vec{E} \times \vec{B}$ (D) none of these

Q5. Select wrong statement from the following for EMW -

(A) are transverse

(B) travel with same speed in all medium

(C) travel with the speed of light

(D) are produced by accelerating charge

Q6. The wave function (in S.I.units) for an electromagnetic wave is given as –

$\psi(x,t) = 10^3 \sin\pi(3\times10^6 x - 9\times10^{14} t)$

The speed of the wave is –

(A) 9×10^{14} m/s (B) 3×10^{8} m/s

(C) 3×10^{6} m/s (D) 3×10^{7} m/s

Q7. The electric field part of an electromagnetic wave in a medium is represented by $E_x = 0$;

$E_y = 2.5\ \frac{N}{C}\cos\left[\left(2\pi\times10^6\ \frac{rad}{m}\right)t - \left(\pi\times10^{-2}\ \frac{rad}{s}\right)x\right]$;

$E_z = 0$. The wave is :

(A) Moving along – x direction with frequency 10^6 Hz and wave length 200 m.

(B) Moving along + y direction with frequency $2\pi \times 10^6$ Hz and wave length 200 m.

(C) Moving along + x direction with frequency 10^6 Hz and wave length 100 m.

(D) Moving along + x direction with frequency 10^6 Hz and wave length 200 m.

Q8. An electromagnetic wave with frequency ω and wavelength λ travels in the + y direction. Its magnetic field is along-x axis. The vector equation for the associated electric field (of amplitude E_0) is :-

(A) $\vec{E} = E_0\cos\left(\omega t - \frac{2\pi}{\lambda}y\right)\hat{x}$

(B) $\vec{E} = -E_0\cos\left(\omega t + \frac{2\pi}{\lambda}y\right)\hat{x}$

(C) $\vec{E} = -E_0\cos\left(\omega t + \frac{2\pi}{\lambda}y\right)\hat{z}$

(D) $\vec{E} = E_0\cos\left(\omega t - \frac{2\pi}{\lambda}y\right)\hat{z}$

Q9. Which of these statements correctly describes the orientation of the electric field $(\vec{E})$, the magnetic field $(\vec{B})$ and velocity of propagation $(\vec{v})$ of an electromagnetic wave?

(A) $\vec{E}$ is perpendicular to $\vec{B}$; $\vec{v}$ may have any orientation relative to $\vec{E}$.

(B) $\vec{E}$ is perpendicular to $\vec{B}$; $\vec{v}$ may have any orientation perpcndicular to $\vec{E}$.

(C) $\vec{E}$ is parallel to $\vec{B}$; $\vec{v}$ is perpendicular to both $\vec{E}$ and $\vec{E}$.

(D) Each of the three vectors is perpendicular to the other two.

Q10. The rms value of the electric field of the light coming from the sun is 720 N/c. The average total energy density of the electromagnetic wave is :

(A) 4.58×10^{-6} J/m^3

(B) 6.37×10^{-9} J/m^3

(C) 81.35×10^{-12} J/m^3

(D) 3.3×10^{-3} J/m^3

Q11. During the propagation of electromagnetic waves in a medium :

(A) Electric energy density is equal to the magnetic energy density

(B) Both electric magnetic energy densities are zero

(C) Electric energy density is double of the magnetic energy density

(D) Electric energy density is half of the magnetic energy density.

Q12. The AM range of radiowaves have frequency –

(A) less than 30 MHz

(B) more than 30 MHz

(C) less than 20000Hz

(D) more than 20000Hz

Q13. The condition under which a microwave oven heats up a food item containing water molecules most efficiently is :-

(A) Infra-red waves produce heating in a microwave oven.

(B) The frequency of the microwaves must match the resonant frequency of the water molecules.

(C) The frequency of the microwaves has no relation with natural frequency of water molecules.

(D) Microwaves are heat waves, so always produce heating.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	C	C	A	C	C	D	C	D	B
11	**12**	**13**	**14**	**15**					
D	C	D	B	D					

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	B	D	C	B	B	D	D	D	A
11	**12**	**13**							
A	A	B							

-: Important Notes & Formulas :-